

Experiment 14: House Price Prediction**Aim**

To implement a **House Price Prediction model** using an appropriate machine learning algorithm.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the house price dataset.

Step 3: Handle missing values in the dataset.

Step 4: Select the relevant input features and target price.

Step 5: Convert categorical features into numerical values if required.

Step 6: Split the dataset into training and testing sets.

Step 7: Create a Linear Regression model.

Step 8: Train the model using the training data.

Step 9: Predict house prices for the test data.

Step 10: Calculate and display the R^2 score.

Program

```
# Experiment 14 - House Price Prediction

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("Housing.csv")

data = data.dropna()
data = pd.get_dummies(data, drop_first=True)

X = data.drop("price", axis=1)
y = data["price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Prices:", y_pred)
print("R2 Score:", r2_score(y_test, y_pred))
```

Output

```
*** Predicted Prices: [5164653.90033967 7224722.29802166 3109863.24240338 4612075.32722559
3294646.25725955 3532275.09556558 5611774.56836476 6368145.98732718
2722856.95689985 2629405.61585782 9617039.5031558 2798087.30447888
3171096.76847063 3394639.09125528 3681088.65424275 5263187.74621485
3035963.47612384 4786122.8004005 4349551.9200572 3572362.09930451
5774875.21395649 5886993.57919883 2730836.19518458 4727316.47323635
5244847.52716801 7555324.21605603 3220790.84680268 5191898.79934206
8143726.91009786 3398814.09825035 6490693.05027925 3315105.9074781
6708457.36761326 4201738.21071676 3557571.06735186 5836974.50478626
4808660.67448475 4362878.73613261 3191242.95701508 4596554.93225238
4566042.86048411 3517779.52374148 7205844.79365835 3983597.27861103
3749338.70271056 4274731.09125895 6757442.10783741 4037320.43665851
3769334.90397126 3417627.44377156 7268416.67644611 2802534.35431384
4341750.61420452 4516422.15345642 3679997.65037847 2678959.06481152
7498029.17782458 2920366.45493167 4239582.05960181 2766467.48467033
4841309.85933272 3584359.56014922 5142140.8489758 4221869.20996263
4698408.50824081 4662380.98564312 7177841.51216424 3452198.9138376
5902436.91361412 6198094.31822813 4773218.99689077 5178933.32585698
4490410.9610656 7968737.41567514 3519394.94792195 5501905.04909554
3947055.15571395 4714056.36707951 4792291.26764381 4250596.74659378
7718149.1935735 4053241.11775758 6453918.60757726 5355571.58680279
2755188.29175269 6796008.09793359 2607277.38176331 3679738.85761987
7968276.12638738 8078480.59500816 3266213.77309707 5897888.53031042
3569461.07439185 3868964.9803252 7584429.53124892 4740586.43771692
5139372.69032253 6423485.10228602 4914078.82258489 5982087.31289316
3855027.3219428 6420308.27282364 3749108.51314829 5998642.37204432
4973331.10029253 4336650.86896384 7059062.72274841 6398940.86831986
6363889.85312644]
R2 Score: 0.6529242642153184
```

Result

Thus, the **House Price Prediction model** was successfully implemented using **Linear Regression**, and the house prices were predicted for the test data.